DATA VORTEX | ROUND 3 ANALYTICAL REPORT

Social Engine: Live Conversation Analysis

Topic :

Public Reaction to a New Pricing or Subscription Model

Case Study :

Meta One Subscription Model

Participant details:

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Team Size : 1

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Analysis Period : 15-20 September 2026

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1. Abstract

This report presents a social conversation analysis for Round 3 of DATA VORTEX A' 26. The assigned topic was Public Reaction to a New Pricing or Subscription Model, studied through the Meta One subscription model as a case study.

A self-collected dataset of 166 publicly accessible records was compiled from recently published news sources and verified public Reddit discussions between 15 and 20 September 2026. The sentiment classification model developed during Round 2, based on TF-IDF and Multinomial Naive Bayes, was reused to analyse the

newly collected data.

The analysis examines sentiment distribution, daily activity, sentiment shifts, discussion topics, entities and temporal activity patterns. The results identify 16 September 2026 as the highest-activity day, with 82 records and an activity Z-score of 1.649. Subscription and plan-related discussions were the most frequently identified topic.

2. Problem Statement

Subscription-based pricing models can generate rapidly changing public conversations across online platforms and news sources. These conversations may contain positive reactions, negative reactions, questions about pricing, discussions about premium features and concerns about the value of paid services.

The objective of this analysis is to apply an existing NLP model to newly collected public data and identify sentiment changes, activity patterns, discussion topics, entities and possible temporal triggers surrounding a new subscription model.

3. Objectives

The main objectives were:

1. Collect recently published public data relevant to the assigned topic.

2. Apply the Round 2 sentiment classification model to the newly collected data.

3. Analyse sentiment distribution over time.

4. Identify significant changes in sentiment composition.

5. Detect periods of unusually high activity.

6. Identify major discussion topics.

7. Identify frequently mentioned entities.

8. Examine the temporal relationship between activity changes and the evolving discussion.

4. Data Collection Method

The assigned topic was Public Reaction to a New Pricing or Subscription Model. Meta One was selected as the concrete case study for analysing this type of public reaction.

The collected dataset combines:

• **News:** recently published relevant articles collected through Google News RSS.

• **Reddit:** relevant publicly accessible and verified Reddit discussion pages.

The observation period covered **15 September 2026 to 20**

September 2026.

The final unified dataset contains:

• 166 total records

• 157 News records

• 9 Reddit records

Each record was structured using fields including platform, source, content type, text, publication timestamp, source URL and collection method.

The collection approach focused on relevant discussions and publications concerning Meta One, subscription plans, pricing, premium features and related reactions.

5. Dataset Overview

Attribute	Value
Total records	166
News records	157
Reddit records	9
Observation period	15 - 20 Sep 2026
Case study	Meta One
Sentiment model	TF-IDF + Multinomial Naive Bayes

The dataset contains recently published public information and therefore represents a collected public-source sample, rather than the complete online conversation surrounding Meta One.

6. NLP Methodology

The Round 2 sentiment model was reused without retraining.

Processing Pipeline

Collected Public Data

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Text Preprocessing

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TF-IDF Transformation

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Round 2 Multinomial Naive Bayes Model

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Sentiment Prediction

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Temporal + Topic + Entity Analysis

The model produces three sentiment classes:

- Negative
- Neutral
- Positive

The Round 3 sentiment labels are model predictions and should not be interpreted as manually verified ground-truth labels.

7. Sentiment Analysis

The daily sentiment distribution shows that positive sentiment was the dominant predicted sentiment during the main observation period.

Date	Negative	Neutral	Positive
15 Sep	6	12	36
16 Sep	4	18	60
17 Sep	2	3	13
18 Sep	1	1	2
19 Sep	1	2	3
20 Sep	0	1	1

On 16 September, the dataset recorded the largest daily volume with 82 records.

The later dates contain substantially fewer observations, so percentage-based sentiment changes on those dates should be interpreted cautiously.

8. Activity Analysis

Daily activity was analysed using the number of collected records and an activity Z-score.

Key observation

16 September 2026

• Records: 82

• Activity Z-score: 1.649

• Spike threshold: 1.5

• Detected: Yes

Therefore, 16 September was identified as the only detected activity spike during the observation period.

Important methodological note

The dataset does not contain usable numeric engagement values. Therefore, this result represents an activity **spike based on record volume**, not a social-media engagement spike.

9. Sentiment Shift Analysis

The daily sentiment-composition changes were examined using percentage-point differences.
The largest observed movement occurred on **18 September 2026**.

Sentiment	**Change**
Negative	+13.89 pp
Neutral	+8.33 pp
Positive	-22.22 pp

The decrease in positive sentiment share coincided with increases in negative and neutral shares.

This should be interpreted as an observed temporal sentiment shift, not proof that a specific event caused the change.

10. Topic Analysis

Keywords-based topic analysis identified the following major discussion theme:

Topic	Records
Subscription / Plans	126
AI / AI Tools	75
Pricing/ Cost	40
Premium Features	40
Monetization	28
Creator Tools	19
Advertising	3

• Subscription / Plans was the most frequently identified topic, appearing in 126 records.

• AI and AI-tool discussions were the second most frequently identified theme with 75 records.

These topic categories are not mutually exclusive. A single record may contain keywords associated with multiple topics.

11. Entity Analysis

The most frequently identified entities were :

Entity	Records
Meta	164
Meta One	162
Instagram	43
Whatsapp	39
Facebook	35
Meta AI	10

Meta and Meta One were the dominant entities in the collected dataset.

Instagram, WhatsApp and Facebook appeared as related platform entities, reflecting the broader ecosystem associated with the subscription model.

12. Trigger / Event Explanation

16 September 2026 was identified as the main activity spike in the observation period.

The day contained **82 records**, producing an activity Z-score of **1.649**, which exceeded the defined threshold of 1.5.

The collected discussion during this period was associated with several recurring themes:

- Subscription plans
- Pricing
- Premium features
- AI-related tools

• Meta platform rollout

The concentration of these discussions coincided temporally with the detected activity spike.
However, the analysis does not establish a causal relationship between any particular event and the observed activity increase.

13. Key Findings

The analysis produced the following major findings:

1. Positive sentiment was the dominant predicted sentiment across the collected dataset.

2. 16 September 2026 recorded the highest activity volume with 82 records and was the only detected activity spike.

3. Subscription / Plans was the most frequently identified discussion topic, with 126 records.

4. AI / AI Tools was the second most frequently identified topic, with 75 records.

5. Meta and Meta One were the most frequently mentioned entities.

6. 18 September showed the largest observed sentiment-composition movement, with positive sentiment decreasing by 22.22 percentage points.

7. Later dates contained substantially fewer observations, requiring caution when interpreting percentage-based changes.

14. Limitations

The analysis has several limitations:

• The sentiment labels are predictions generated using the Round 2 TF-IDF and Multinomial Naive Bayes model.

• TF-IDF-based sentiment classification may have difficulty with sarcasm, implicit sentiment and context-dependent language.

• Topic and entity identification uses keyword-based matching.

• Topic categories are therefore not mutually exclusive.

• The collected public sources do not represent the complete online conversation.

• Later dates contain fewer records than the main observation period.

• Numeric engagement data was not available in the unified dataset.

• The detected activity spike represents record volume rather than engagement.

• Temporal association does not establish causality.

15. Conclusion

This Round 3 analysis demonstrates the reuse of the Round 2 NLP model on a newly collected public dataset related to **public reaction to a new pricing or subscription model.**

Using **166 collected records**, the analysis examined sentiment, activity patterns, sentiment shifts, topics and entities surrounding the Meta One case study.

The analysis identified **16 September 2026** as the highest-activity day, with **82** records and an activity Z-score of **1.649**. Subscription and plan-related discussions were the most frequently identified theme, while Meta and Meta One were the most frequently mentioned entities.

A substantial sentiment-composition movement was observed on 18 September, including a 22.22 percentage-point decrease in positive sentiment share. Because the number of observations varied considerably across dates, these changes were interpreted alongside activity volume.

Overall, the project demonstrates how an existing NLP model can be applied to newly collected public data to monitor **sentiment, activity patterns, discussion themes and entities over time.**